

Development of a PM2.5 Measurement Device Using Dust Sensors for Air Quality Monitoring

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Abstract— This research focuses on the development of an advanced air purifier system integrated with PM2.5 detection, UV-C light sterilization, and real-time monitoring through an IoT-based interface. The system utilizes a PM2.5 sensor to detect airborne particulate matter and activates the air purifier and UV-C light when the air quality exceeds predefined thresholds. Additionally, the system is designed to display PM2.5 values on an LCD screen and send notifications to a mobile application (LINE) to inform users about air quality conditions. Prototype testing was conducted in a 6 x 12-meter room using hairspray to simulate PM2.5 particles. The results showed that the air purifier activated successfully in response to elevated PM2.5 levels, displaying real-time data and sending alerts with messages such as "Danger" when air quality worsened. This study demonstrates the feasibility of combining IoT technology and real-time air quality monitoring to enhance environmental health management in indoor spaces. The proposed system provides a user-centric solution that not only improves air quality but also ensures immediate user awareness and intervention.

Keywords— Air Purifier, PM2.5 Detection, UV-C Sterilization, IoT, Real-Time Monitoring, LINE Notification, Environmental Health, Air Quality Management.

I. INTRODUCTION

In recent years, Thailand, along with many countries worldwide, has been facing a widespread crisis concerning the hazardous levels of fine particulate matter smaller than 2.5 microns, commonly known as PM2.5 (Particulate Matter 2.5) [1]. This particulate matter is imperceptible to the naked eye, which leads to a lack of awareness about its severe health implications [2]. PM2.5 primarily originates from the combustion of machinery, particularly the engines of both new and old vehicles, which release significant amounts of particulate matter during periods of heavy traffic, especially during the morning and evening rush hours. Such pollution is most prevalent in winter when the air tends to be stagnant and dry, preventing the dust from rising. Additionally, Thailand is currently experiencing challenging atmospheric conditions, including poor airflow caused by high-rise buildings that obstruct wind patterns and contribute to the deterioration of air quality [1], [3]. The widespread dust from construction activities further exacerbates this issue, significantly impacting urban air quality.

The inhalation of PM2.5 poses serious health risks, particularly to the respiratory system [4]. Exposure can lead to irritation of the airways, causing symptoms such as nasal discomfort, coughing, sneezing, and phlegm production [2]. Prolonged exposure to these fine particles has been linked to more severe health conditions, including sudden heart failure

and, in the most extreme cases, lung cancer [5]. The urgency of addressing this environmental health crisis underscores the need for comprehensive research and prompt action to mitigate its adverse effects on public health [6], [7].



Fig. 1 The image depicting a city in Thailand with visible air pollution, highlighting the effects of PM2.5

According to the World Health Organization (WHO), the safe threshold for particulate matter (PM2.5) in the air is 25 micrograms per cubic meter [1]. Levels exceeding this concentration are considered hazardous to human health. In Thailand, the permissible limit for PM2.5 is set at 50 micrograms per cubic meter [8]. Given the aforementioned issues, it is undeniable that the installation of air purifiers within buildings or residences is crucial for monitoring the average concentration of PM2.5 in the air. This technology plays a key role in mitigating the health risks posed by fine particulate matter [9]. The development of PM2.5 measuring devices relies on dust sensors, which operate based on the principle of laser light transmission [10]. The sensor emits a laser beam through airborne dust particles, which are drawn into the device by a small fan. As the dust particles pass through the laser light, the device detects their presence and quantifies their concentration [11]. This allows for the accurate measurement of fine particulate matter, specifically PM2.5, providing valuable data for air quality monitoring and health protection.

II. LITERATURE REVIEWS

A. Human-Centered Design

Human-Centered Design (HCD) is a modern approach to design that places the needs, behaviors, and perspectives of users at the core of the creative process [12]. The primary focus of HCD is to solve problems from the genuine viewpoints and needs of individuals, resulting in designs that are both aesthetically pleasing and functionally effective in meeting user demands [12], [13]. For example, in architectural design, while aesthetics is an important consideration, the Human-Centered Design approach emphasizes understanding the behaviors, needs, and challenges faced by the users of a given space [14]. Architects, therefore, must take into account who the end-users are, their behavioral patterns, and the issues they have previously encountered, in order to create a design that enhances the users' experience and satisfaction. The principles of Human-Centered Design can be applied across various fields and industries, including architecture, interior design, product development, and even in the startup business sector [12]. Many startups have adopted HCD principles to innovate and create business models that more accurately address the specific needs and desires of consumers. This approach ensures that the design process is user-driven, promoting solutions that cater directly to the demands of the people they serve [14].

B. PM2.5 Detection

An environmental monitoring system for agricultural greenhouses was developed, utilizing IoT and sensors with communication via Zigbee technology [15]. An automated fish feeding system was created with NodeMCU V.2, offering manual and automated modes, with data exchange via a Blynk server [16]. A system for measuring PM2.5 and PM10 and analyzing climate change was developed using electron microscope images for particle size identification and wireless data transmission. A real-time PM2.5 measurement device using NodeMCU ESP8266 and a laser-based sensor was designed, providing data and alerts via the LINE application. Research explored the biological effects of PM2.5, linking it to oxidative stress and adverse pregnancy outcomes [17]. The efficacy of UV-C radiation in pathogen destruction was highlighted. A mobile UV-C disinfection unit was developed for COVID-19 response, improving safety and reducing contamination. An air purifier with ozone, UV-C, and HEPA filters was designed, effectively reducing PM2.5 and pathogens. An overview of LCD technology used in embedded systems was provided, and a system for managing project data and evaluations using Google Apps for Education and LINE Notification API was applied to streamline workflows [15].

III. PROPOSED METHOD

A. Methodology: Empathy Map Development

The development of an air purifier that effectively meets user needs begins with a deep understanding of the users' requirements and challenges. This approach ensures the design of a product that addresses users' concerns and provides practical solutions. Utilizing the Empathy Map as a tool for data collection through surveys and interviews allows for a comprehensive understanding of user needs. The

resulting Empathy Map summarizes key issues related to air purifier usage, providing a valuable framework for design decisions. The data collected revealed that users are particularly concerned about their health, especially those with allergies, and fear the potential health impacts of exposure to PM2.5 particulate matter. Additionally, concerns regarding electricity costs and maintenance expenses were prevalent. Users expressed uncertainty about when to activate the air purifier, when to change filters, and were unaware of the current PM levels in their environment. This lack of clarity in usage instructions creates inconvenience and confusion for users. In terms of users' thoughts and feelings, the data indicated that health concerns and financial worries were significant factors influencing their experience with air purifiers. Furthermore, users desired the ability to view real-time PM2.5 readings and track historical data. They also requested a notification system to alert them when PM levels exceed recommended thresholds or when filter replacement is necessary. The findings emphasize the need for an air purifier design that goes beyond simply filtering particulate matter. It is crucial to integrate features that enhance user convenience, such as real-time data displays, automatic operation, and notifications for filter changes and PM level fluctuations. Such features would provide users with clear, actionable information, thereby improving their overall experience with the product. Designing an air purifier that incorporates these features will add significant value by addressing both health concerns and cost-related issues. This approach highlights the importance of developing user-centric products that prioritize not only technological performance but also user experience. In conclusion, the design of air purifiers should be informed by a comprehensive understanding of user needs and preferences, ensuring that the product effectively contributes to both health protection and user convenience.

Empathy Map

Say & Do

- Typically, users tend to turn on the air purifier upon returning home, although sometimes they forget to do so.
- The air purifier is usually operated from 6 p.m. until morning on weekdays. However, on weekends, it is often left running throughout the day.
- The current device being used lacks a notification system or display for PM2.5 readings.

Think & Feel

- Users express concern about their health, particularly due to allergies.
- There is a significant fear regarding the potential health consequences of exposure to particulate matter (PM), such as lung cancer.
- Concerns also arise about electricity costs and ongoing maintenance expenses.
- A reasonable price for an air purifier is perceived to be between 4,000 and 5,000 Baht.

See

- Users generally display a relaxed demeanor but express concerns when discussing the issue of air pollution.
- They respond clearly to inquiries and offer insightful suggestions.

- Users tend to be knowledgeable and appear modern, often working in professional environments such as universities and companies.

Hear

- Users express reluctance to manually turn on the air purifier.
- They desire an all-in-one system that simplifies operation.
- Forgetfulness is a recurring theme, with users often forgetting to carry their mobile phones.
- Users express interest in real-time notifications about air quality levels.
- The device size is expected to match the actual usable space, with the user also forgetting to change the filter.

Pain

- There is uncertainty regarding when to activate the air purifier.
- Users are unsure when to change the filter.
- They lack knowledge of the current PM2.5 levels in their environment, leading to frustration and inconvenience.

Gain

- The ideal air purifier should feature automatic operation upon arrival home.
- Real-time PM2.5 readings should be clearly displayed.
- A notification system would alert users when PM levels exceed safe limits or when the filter needs replacement.
- The device should be capable of disinfecting both airborne pathogens and particulate matter, including tuberculosis and influenza viruses, thus enhancing its utility.

B. Customer Journey Map

1. Awareness

In the initial stage of the customer journey, the user begins by searching for solutions to manage PM_{2.5} particulate matter, utilizing relevant search terms related to air quality and the risks associated with airborne pollutants. At this stage, users often encounter a wide range of information that may not directly address their specific needs, leading to frustration and confusion. The lack of clarity in available resources can hinder the decision-making process, delaying the progression to the next stage.

2. Consideration

As users develop an interest in addressing their concerns with an air purifier, they begin to explore options more thoroughly. This typically involves researching products through e-commerce websites and social media platforms. However, during this phase, users face challenges in navigating the complexity of product choices and differentiating between various models, leading to uncertainty. Users may feel a lack of confidence in their ability to make an informed decision due to the overwhelming number of choices and insufficient product information.

3. Purchase

Once users decide on a suitable air purifier, they proceed with the purchase through an e-commerce platform. The payment process, however, often introduces obstacles such as complexity or inconvenience, which may result in a less-than-optimal user experience. These issues can cause dissatisfaction and frustration, underscoring the importance of streamlining the payment process for improved user convenience. Ensuring a smooth, efficient transaction is critical to reinforcing the positive customer experience.

4. Onboarding

After purchasing the air purifier, users enter the onboarding phase, where they begin to set up and use the product. Challenges often arise when users struggle with understanding the initial setup procedures or configuration of the device. This lack of clarity can lead to feelings of stress or confusion, particularly if the user is unfamiliar with the product's features. Providing clear, user-friendly instructions and offering automated setup features can alleviate these concerns and enhance the user's experience in the onboarding process.

5. Advocacy

Satisfied users, having had a positive experience with the product, are likely to recommend it to others, either through word-of-mouth or by sharing their experiences on social media platforms. However, if users do not feel adequately acknowledged or rewarded for their advocacy, they may be less inclined to continue promoting the product. Therefore, implementing referral programs or recognizing user contributions can incentivize further advocacy, creating a cycle of positive reinforcement and customer loyalty.

C. Ideate



Fig. 2 Ideate

Fig.2 shows, the design concept for an advanced air purifier system integrates various features aimed at enhancing user experience and promoting better air quality management. The core attributes of the system, as illustrated in the Idea Board, focus on both functionality and user convenience, ensuring that the product meets the needs of consumers across different environments, such as homes and offices. The air purifier is priced between 3,000 to 5,000 Baht, which positions it as an affordable yet efficient option for consumers seeking to manage particulate matter (PM) levels. The device is designed to offer a variety of practical benefits, including a built-in warning system for PM particle levels, allowing users to be alerted when air quality falls below acceptable thresholds. Additionally, the system features an indicator to remind users when the filter needs to be replaced, which supports effective long-term maintenance. The design of the

device is aesthetically pleasing, with a sleek and simple design, available in white, black, gray, and blue colors, making it versatile and adaptable to various interior designs. It is constructed from high-quality, durable materials that ensure the longevity of the product. The air purifier is also portable, allowing easy relocation for optimal air quality management in different spaces.

Notable features of the device include the ability to operate using electricity or a power bank, providing flexibility in terms of power sources. It also incorporates a wireless charging function, adding to the convenience for users. Furthermore, the air purifier is capable of not only purifying the air but also eliminating germs, enhancing its role in promoting healthier living environments. The system's user-friendly interface includes a power indicator light that alerts users to the status of the filter, helping to maintain efficiency and optimize performance. Overall, the design concept addresses the growing concerns about air quality and health by offering a comprehensive, reliable, and adaptable solution for individuals and businesses alike.

D. Prototype

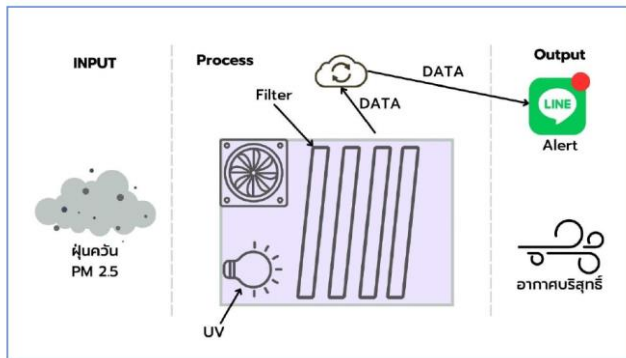


Fig.3 Conceptual Framework

The conceptual framework for the air purifier system integrates various components that work together to monitor, filter, and alert users regarding particulate matter (PM2.5) levels in the air. This system functions through a streamlined process that incorporates both data collection and real-time feedback, aimed at enhancing user awareness and providing actionable insights for better air quality management. Input, the system begins by receiving input in the form of airborne particulate matter (PM2.5), which is measured in real-time. These fine particles, often invisible to the naked eye, are detected by the system's sensors, which continuously monitor the surrounding environment for air quality. The input data also includes environmental factors such as humidity, temperature, and UV radiation, which could influence the purifier's operation. Process, once the input data is received, the system processes the information using an integrated filtering mechanism. The primary processing unit involves a filter that captures particulate matter, effectively removing harmful particles from the air. This process is assisted by UV light, which further purifies the air by eliminating germs and bacteria, ensuring that the environment is both free from airborne pollutants and safer for the users. The data is simultaneously collected and processed, enabling real-time monitoring of air quality. Output, the processed data is then transmitted to the cloud, where it is stored and analyzed for

future reference. The system outputs critical information about the current air quality, including real-time data on PM2.5 levels. The output is sent to the user via a mobile application, such as LINE, which provides an alert notification when PM levels exceed the safe threshold or when other significant changes in air quality occur. This notification system allows users to take immediate action, such as activating the air purifier or adjusting their environment to improve air quality.

The conceptual framework highlights the importance of seamless interaction between sensors, data processing, and real-time user feedback. By delivering timely alerts through familiar platforms like LINE, the system ensures that users remain informed about the air quality in their environment, ultimately leading to healthier living spaces. This proactive approach encourages users to take action before the air quality reaches harmful levels, thus mitigating the adverse effects of long-term exposure to PM2.5. The integration of data collection, processing, and real-time notifications into a unified system provides a comprehensive solution for managing air quality. This framework underscores the potential of IoT-based air purifiers in contributing to healthier indoor environments and the advancement of air quality monitoring technologies.

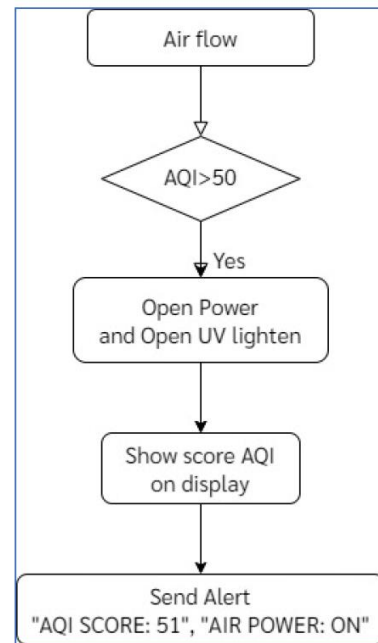


Fig.4 System flowchart

Air Flow Detection

The system begins by detecting the airflow, which is an essential component in determining whether the air quality requires purification. If the air quality index (AQI) is above a threshold of 50, the system proceeds to the next steps.

AQI Measurement

The system measures the Air Quality Index (AQI), which is a numerical value that represents the concentration of particulate matter (PM) in the air. The AQI is compared against a threshold value of 50.

Condition

If $AQI > 50$, proceed to the next step.

Activate Air Purifier and UV Light

If the AQI exceeds 50, the air purifier is activated to clean the air, and the UV light is turned on to sterilize the environment

and eliminate germs or bacteria. This step ensures that the air is purified and safe for the users.

Display AQI Score

The system then displays the current AQI score on the screen, allowing users to see the real-time air quality. This helps users to assess the condition of the air and make decisions accordingly.

Send Alert

Finally, the system sends an alert to the user, displaying the message: "AQI SCORE: 51", indicating the air quality status, and "AIR POWER: ON", signaling that the air purifier is activated.

AQI Calculation

The AQI can be derived from the concentration of particulate matter (PM2.5) in the air using the following general equation:

$$AQI = \frac{PM2.5 \times (50 - 0)}{PM2.5_{\max} - PM2.5_{\min}} + 0$$

where

- $PM2.5$ is the concentration of particulate matter in micrograms per cubic meter.
- $PM2.5_{\max}$ and $PM2.5_{\min}$ are the maximum and minimum permissible PM2.5 concentration values, respectively.

this process ensures that the air purifier is automatically activated when air quality falls below acceptable levels, and alerts the user in real-time, thereby maintaining a healthier environment.

E. Air purifier circuit design

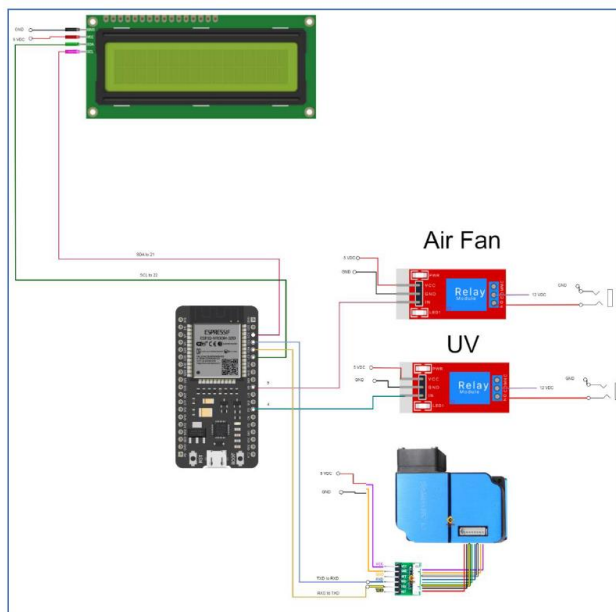


Fig. 5 Automatic air purifier circuit type

Fig. 5 shows the simulation of the air purifier circuit involves connecting an ESP32 board to an air intake fan, a UV-C light bulb, and a PM2.5 dust sensor. The system is designed to display the detected PM2.5 values on an LCD screen. Upon detecting particulate matter in the air, the system activates the intake fan and the UV-C light bulb to improve air quality by filtering out harmful particles and eliminating airborne pathogens. This circuit integration allows for real-time monitoring of air quality, with the ESP32 board serving as

the central control unit for sensor data collection and device operation.

F. Air Purifier Model Overview

The air purifier model is designed with several key components that facilitate its operation. At the top of the device, an intake fan is positioned to draw in air for purification. The system also includes a UV-C light bulb, which is used to sanitize the air. A display screen is integrated into the model, showing the current PM2.5 levels as numerical values. Additionally, the status of the air quality is indicated by a message: when the PM2.5 value is below 51, the message displayed reads "Good," whereas values greater than or equal to 51 trigger the message "Not Good."

Around the device, ventilation openings allow for proper air circulation. Inside the unit, both a UV-C light bulb and an air filter are installed to effectively purify the air. An example of the air purifier model is shown in Figure 6.

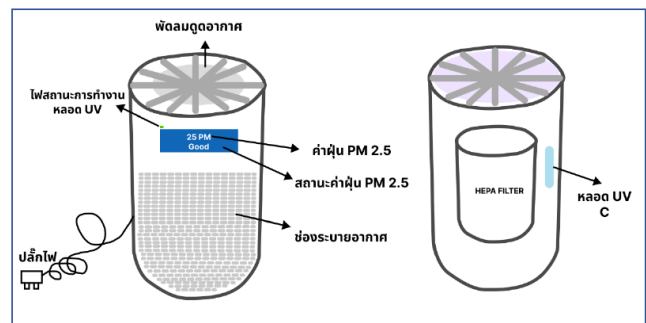


Fig. 6 Air purifier model image

G. Notification via LINE Application



Fig. 7 Notification via LINE Application

The system is designed to send a notification through the LINE application when the PM2.5 sensor detects particulate matter levels exceeding 51. The notification includes the measured PM2.5 value and provides information on the operational status of the air purifier. This feature ensures that users are promptly informed about the air quality and the status of the air purification process. An example of the notification simulation through the LINE application is illustrated in Figure 7.

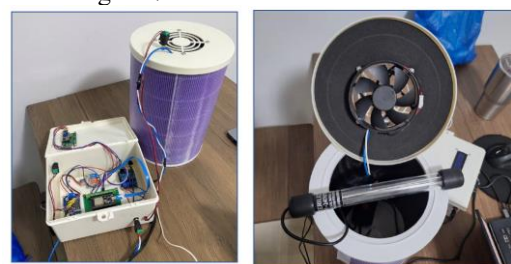


Fig. 8 Prototype

IV. EXPERIMENTAL RESULTS

A. Prototype Testing Process: Human-Centric Design Framework

In the prototype testing phase, it is imperative to follow the problem framework defined in the previous stages, as it serves as the foundation for achieving the primary objective: addressing the identified problem. Simultaneously, the design process follows a Human-Centric Design approach, which prioritizes the end-users throughout the entire design process. Therefore, during the prototype testing phase, it is essential to actively involve users, collecting their feedback and suggestions. This iterative process of engaging with real users ensures that the prototype is refined and improved based on practical insights, ultimately leading to a more user-aligned design in subsequent iterations.

B. Prototype Functionality Testing: First Trial

In the first trial of the prototype testing, the air purifier system was placed in a controlled environment with a room size of 6 x 12 meters. The objective was to evaluate the performance of the sensor by simulating PM2.5 dust levels using hairspray, which was sprayed at a distance of approximately 30 cm from the sensor. The expected outcomes included the activation of the air purifier and the UV-C light, the display of the detected PM2.5 values on the LCD screen, and the transmission of a notification to the LINE application.

Test Procedure:

- **Room Dimensions:** 6 x 12 meters
- **Dust Simulation:** PM2.5 was simulated using hairspray at a distance of 30 cm from the sensor.
- **Expected Results:** Activation of the air purifier, UV-C light, real-time display of PM2.5 levels on the LCD, and an alert sent to the LINE application.

The test successfully demonstrated the functionality of the air purifier system. Upon detecting the simulated PM2.5 levels, the air purifier activated as expected, and the UV-C light was turned on. The PM2.5 value of 200.0 was displayed on the LCD screen along with the message “Danger,” indicating that the air quality exceeded the acceptable threshold. Furthermore, the system sent a notification to the LINE application with the message: “Air Purify: PM2.5 exceeds standard,” accompanied by the detected value, “Air Purify: 2000.” This preliminary testing confirms the system’s capability to respond to real-time air quality changes and effectively alert users through both the visual display and mobile application.

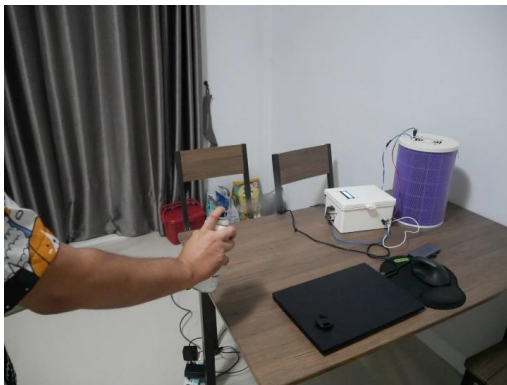


Fig. 9 First Trial



Fig. 10 Test 1. Notification via Line application.

C. Prototype Functionality Testing: Second Trial

In the second trial of the prototype testing, the air purifier system was placed in a room measuring 6 x 12 meters. The objective was to evaluate the performance of the sensor by simulating PM2.5 dust levels using hairspray, sprayed at a distance of approximately 2 meters from the sensor. The expected outcomes included the activation of the air purifier and UV-C light, real-time display of the detected PM2.5 values on the LCD screen, and the transmission of an alert to the LINE application.

Test Procedure:

- **Room Dimensions:** 6 x 12 meters
- **Dust Simulation:** PM2.5 was simulated using hairspray at a distance of 2 meters from the sensor.
- **Expected Results:** Activation of the air purifier and UV-C light, display of PM2.5 levels on the LCD screen, and an alert sent to the LINE application.

The results of the second test showed that the air purifier system functioned as expected. Upon detecting the simulated PM2.5 levels, the air purifier activated, and the UV-C light was turned on. The PM2.5 value of 144.1 was displayed on the LCD screen along with the message “Danger,” indicating that the air quality was above the acceptable threshold. In addition, the system successfully sent a notification to the LINE application with the message: “Air Purify: PM2.5 exceeds standard,” along with the detected value, “Air Purify: 1441.” These results validate the air purifier system’s ability to detect elevated PM2.5 levels and provide timely alerts, ensuring real-time user awareness of air quality conditions.

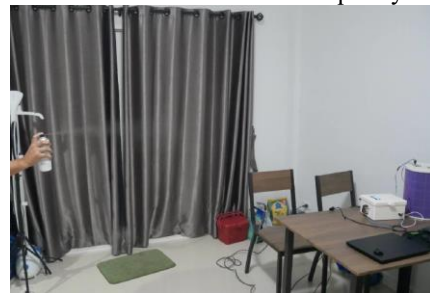


Fig. 11 Second prototype test



Fig. 12 Test 2. Notification via Line application.

V. CONCLUSION

This research demonstrates the successful development and testing of an air purifier system equipped with PM2.5 detection and UV-C light sterilization. The system integrates real-time air quality monitoring, automatic activation of the air purifier and UV-C light, and provides notifications to users via the LINE application, ensuring that they are promptly informed of any significant changes in air quality. Through a series of functional tests conducted in controlled environments, the system effectively responded to PM2.5 concentrations exceeding the threshold by activating the purification process and sending real-time alerts. The system displayed PM2.5 values on an LCD screen, accompanied by status messages such as "Danger" when air quality exceeded acceptable levels. The successful transmission of alerts to the LINE application further enhanced user awareness, providing actionable information for immediate intervention. The results of the prototype testing confirm the feasibility of implementing a human-centric air purification system that not only addresses the issue of air quality but also ensures seamless user interaction and responsiveness. Future work will focus on refining the system for broader real-world applications, optimizing sensor accuracy, and exploring additional features such as integration with other smart home systems. This research contributes to the field of environmental health by providing a practical solution for managing indoor air quality, particularly in urban areas where air pollution is a growing concern. The integration of IoT technology and real-time user feedback represents a significant step forward in creating responsive and adaptive air quality management systems.

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